Two hours

THE UNIVERSITY OF MANCHESTER

INTRODUCTION TO VISUAL COMPUTING (ONLINE)

28 May 2024

14:00 - 16:00

Instructions for candidates

This is an online exam. All questions are answered on Blackboard.

The examination is worth a total of 60 marks.

Additional permitted materials

- The use of calculators is NOT permitted in this exam.
- Rough paper for working out will be provided.
- Students are NOT allowed to bring in their notes.

Instructions	
Total Questions	29
Total Points	60

☐ 11. **Essay: The illumination equation is: [Maths ...** Points: 9

Question

The illumination equation is:

$$I_R = k_{aR} I_{aR} + \frac{I_{pR}}{d'}\left[k_{dR}(\vec{N}\cdot\vec{L}) + k_s(\vec{R}\cdot\vec{V})^n\right]$$

What does the "R" subscript indicate?

There are three "*k*" terms with different subscripts. What do these refer to?

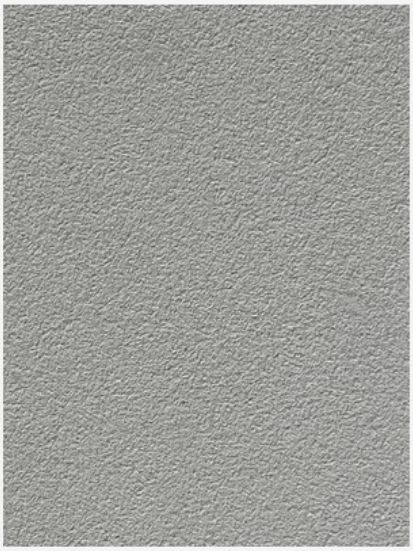
There are two vector expressions: $\vec{N}\cdot\vec{L}$ and $\vec{R}\cdot\vec{V}$. Define the four vectors involved and explain what the expressions achieve.

12. **Essay: You have been asked to create models ...**

Points: 8

Question

You have been asked to create models of the walls like the ones in these two images.





What characteristics of the surfaces of the walls will you use to decide what methods you will use to model and render the walls?

How will you render the surface appearance of the walls?

How will you render the three dimensional structure of the walls (if there is any 3D structure)?

27. **Essay: You want to automatically measure the...** Points: 5

Question

You want to automatically measure the size (area) of the bodies of water fleas in an image as shown below.



You decide to convert the image to greyscale and then threshold to distinguish the fleas from the background. You then think of doing Connected Component Analysis to identify the distinct blobs in the image but realise that the antennae on the fleas (the sticky out bits) will be counted in area. So, you decide to use erosion before doing connected component analysis.

Is erosion a good idea? What will happen to the area of the body? Would the Opening operation be more useful? Explain how the Opening operation works and mention potential problems - how can you remove the antennae if parts of them are quite wide?

28. **Essay: The Prewitt and Sobel operators are c...** Points: 5

Question

The Prewitt and Sobel operators are commonly used to find edges in an image.

Explain why they have a slight smoothing effect as well as finding the edges.

If a Prewitt or Sobel operator is applied to an image that is completely black but with a white square in the middle, the edges detected around the square will be two pixels wide. Explain why that happens.

Prewitt and Sobel kernels come in pairs: one to detect horizontal edges and one to detect the vertical edges. To produce an output showing all strong edges in an image, what is done with the horizontal and vertical intermediate outputs?

It might help your explanation to write example kernels. (You can do this typing them on three lines with spaces between the values.)

29. **Essay: Explain the steps involved in the Mar...** Points: 5

Question

Explain the steps involved in the Marr-Hildreth algorithm for detecting edges in an image.

For each step, give a brief description of what is involved in that step.

If any kernels are involved, briefly mention how they might be calculated.

Can images using an unsigned byte be used for all of the stages?